Exploratory Data Analysis on Electronic game

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Project Background Information

Electronic Game:

operated by computer circuitry. The machines on which electronic games are played include general-purpose shared and personal computers

According to the number of global video game consumers has soared to 3.7 billion.1.5 billion people play on PC. However, this number includes those players who also use consoles and mobile devices.

Electronic games generated US$134.9 billion annually in global sales. And keep growing

Project Objective

- To understand the core characteristics of the video game market, identify key market winners, and determine the critical factors contributing to their success

- Generate insight for future decision-makinggenerate insight for future decision-making

- find out where is the best market

Dataset Description

Dataset

source of dataset :
https://www.kaggle.com/datasets/volodymyrpivoshenko/video-game-sales-dataset

11 Columns, 16000+rows

Data Range: 1980 to 2016

	Rank	Name	Platform	Year	Genre	Publisher \
0	1	Wii Sports	Wii	2006.0	Sports	Nintendo
1	2	Super Mario Bros.	NES	1985.0	Platform	Nintendo
2	3	Mario Kart Wii	Wii	2008.0	Racing	Nintendo
3	4	Wii Sports Resort	Wii	2009.0	Sports	Nintendo
4	5	Pokemon Red/Pokemon Blue	GB	1996.0	Role-Playing	Nintendo
5	6	Tetris	GB	1989.0	Puzzle	Nintendo
6	7	New Super Mario Bros.	DS	2006.0	Platform	Nintendo
7	8	Wii Play	Wii	2006.0	Misc	Nintendo
8	9	New Super Mario Bros. Wii	Wii	2009.0	Platform	Nintendo
9	10	Duck Hunt	NES	1984.0	Shooter	Nintendo

	NA_Sales	EU_Sales	JP_Sales	Other_Sales	Global_Sales
0	41.49	29.02	3.77	8.46	82.74
1	29.08	3.58	6.81	0.77	40.24
2	15.85	12.88	3.79	3.31	35.82
3	15.75	11.01	3.28	2.96	33.00
4	11.27	8.89	10.22	1.00	31.37
5	23.20	2.26	4.22	0.58	30.26
6	11.38	9.23	6.50	2.90	30.01
7	14.03	9.20	2.93	2.85	29.02
8	14.59	7.06	4.70	2.26	28.62
9	26.93	0.63	0.28	0.47	28.31

Variable Identification

Nature	Type	Variable	Data Type
Qualitative	Descriptive	Name	object
	Categorical	Platform	object
		Genre	object
		Publisher	object
Quantitative	Discrete	Rank	Integer
		Year	Float
	Continuous	NA_Sales	Float
		EU_Sales	Float
		JP_Sales	Float
		Other_Sales	Float
		Global_Sales	Float

Data Cleansing

- Check any duplicated Value

```
print (df.duplicated())
print ('the total number of duplicated is', df.duplicated().sum())
```

✓ 0.0s

```
0        False
1        False
2        False
3        False
4        False
         ...
16593    False
16594    False
16595    False
16596    False
16597    False
Length: 16598, dtype: bool
the total number of duplicated is 0
```

Data Cleansing

- Year column has 271 missing values

- Publisher column has 58 missing values

```
print (df.isnull().sum())
missing_value = df[df.isna().any(axis=1)]
print (missing_value)
```

✓ 0.0s

```
Rank              0
Name              0
Platform          0
Year            271
Genre             0
Publisher        58
NA_Sales          0
EU_Sales          0
JP_Sales          0
Other_Sales       0
Global_Sales      0
dtype: int64
```

Data Cleansing

	Rank	Name	Platform	Year	Genre	\
179	180	Madden NFL 2004	PS2	NaN	Sports	
377	378	FIFA Soccer 2004	PS2	NaN	Sports	
431	432	LEGO Batman: The Videogame	Wii	NaN	Action	
470	471	wwe Smackdown vs. Raw 2006	PS2	NaN	Fighting	
607	608	Space Invaders	2600	NaN	Shooter	
...	...	...	...	...	...	

	Publisher	NA_Sales	EU_Sales	JP_Sales	\
179	Electronic Arts	4.26	0.26	0.01	
377	Electronic Arts	0.59	2.36	0.04	
431	Warner Bros. Interactive Entertainment	1.86	1.02	0.00	
470	NaN	1.57	1.02	0.00	
607	Atari	2.36	0.14	0.00	
...	...	...	...	...	
16427	Unknown	0.01	0.00	0.00	
16493	Unknown	0.00	0.01	0.00	
16494	NaN	0.00	0.01	0.00	
16543	NaN	0.00	0.01	0.00	
16553	NaN	0.00	0.01	0.00	

Data Cleansing

Year	Action: Dropna 1,Minimal Impact 2,Time Efficiency 3,Ensuring Accuracy
Publisher	Action: Fillna('Unknown') 1,Data Preservation 2,Easy Tracking

Data Cleansing

```python
df['Publisher'].fillna ('unknown',inplace = True)
df= df.dropna(subset=['Year'])


print (df.info())
```

✓ 0.0s

```
<class 'pandas.core.frame.DataFrame'>
Index: 16327 entries, 0 to 16597
Data columns (total 11 columns):
 #   Column        Non-Null Count  Dtype
---  ------        --------------  -----
 0   Rank          16327 non-null  int64
 1   Name          16327 non-null  object
 2   Platform      16327 non-null  object
 3   Year          16327 non-null  float64
 4   Genre         16327 non-null  object
 5   Publisher     16327 non-null  object
 6   NA_Sales      16327 non-null  float64
 7   EU_Sales      16327 non-null  float64
 8   JP_Sales      16327 non-null  float64
 9   Other_Sales   16327 non-null  float64
 10  Global_Sales  16327 non-null  float64
dtypes: float64(6), int64(1), object(4)
```

Data Cleansing

- Year is fundamentally an integer

```
df['Year']= df['Year'].astype(int)
```

- Avoiding Potential Analytical Errors

```
2   Platform     16327 non-null  object
3   Year         16327 non-null  int64
4   Genre        16327 non-null  object
```

Dataset Overview

- The Substantial Gap Between the Mean and Median
- should be Positive Skewness
- Winner-take-all market
- small number of games pull up the average sales
- The median is a better representation

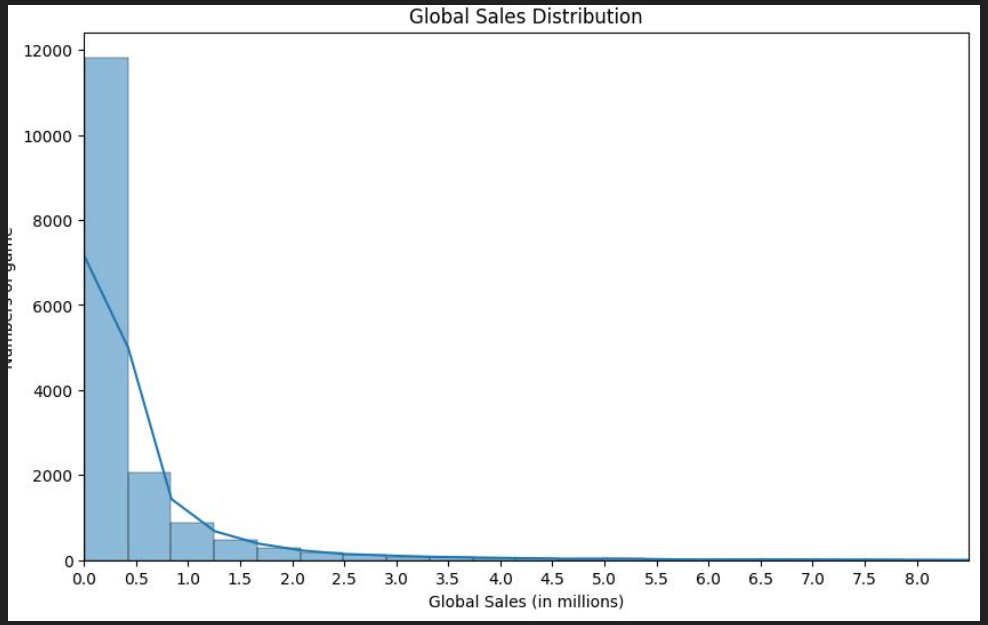
	NA_Sales	EU_Sales	JP_Sales	Other_Sales	Global_Sales
count	16598.000000	16598.000000	16598.000000	16598.000000	16598.000000
mean	0.264667	0.146652	0.077782	0.048063	0.537441
std	0.816683	0.505351	0.309291	0.188588	1.555028
min	0.000000	0.000000	0.000000	0.000000	0.010000
25%	0.000000	0.000000	0.000000	0.000000	0.060000
50%	0.080000	0.020000	0.000000	0.010000	0.170000
75%	0.240000	0.110000	0.040000	0.040000	0.470000
max	41.490000	29.020000	10.220000	10.570000	82.740000

Univariate Analysis

Global Sales Distribution

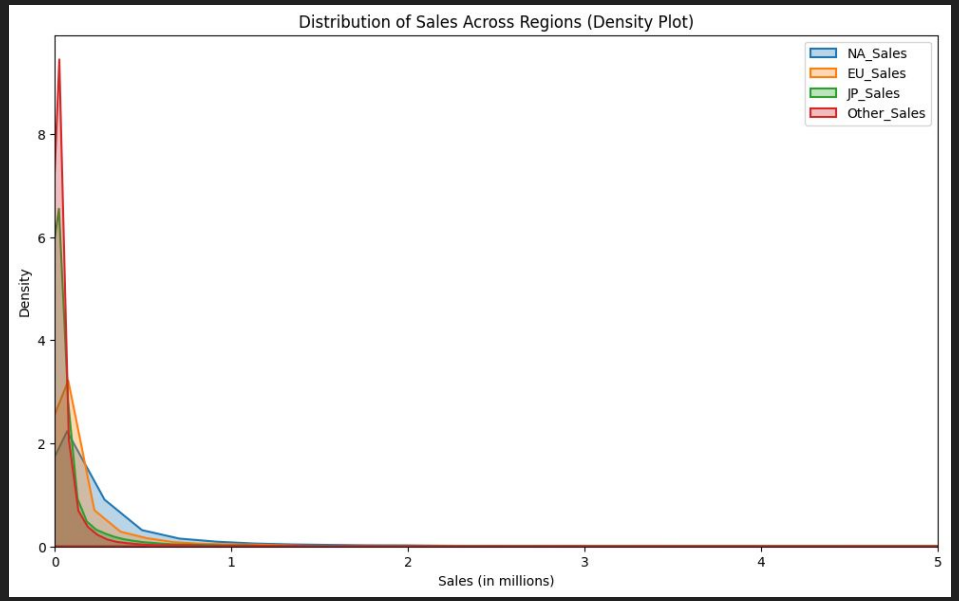
- more than 75% of game sales are below 0.5million

- indicates significant positive skewness and a winner-takes-all market



Distribution of Sales Across Regions

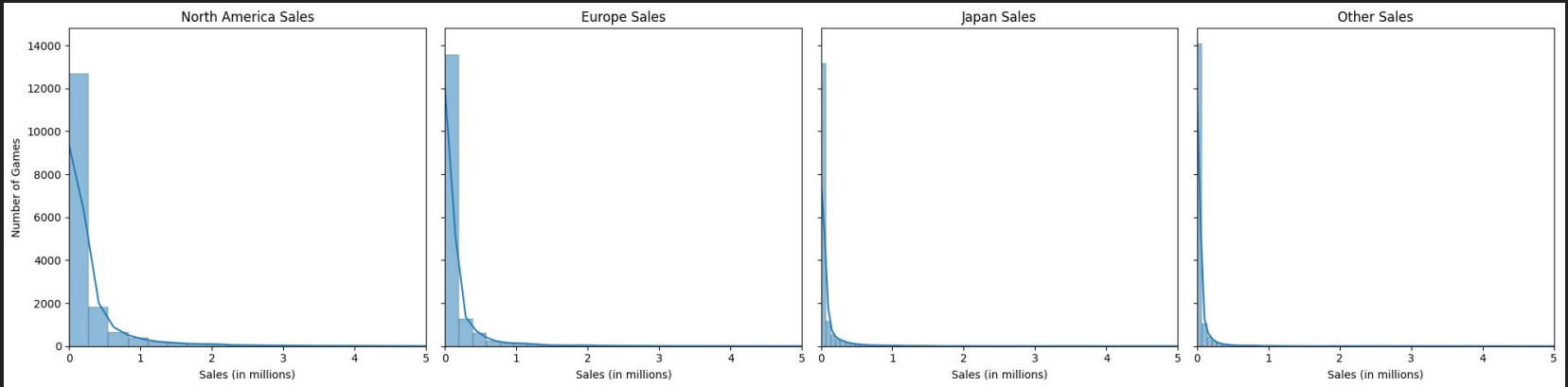
- Still Positive Skewness
- Very Low Median Sales
- NA and EU has similar situation (skewness)



Distribution of Sales Across Regions

- Japan market has lower skewness,High Concentration

- Other regions has highest skewness,High Risk



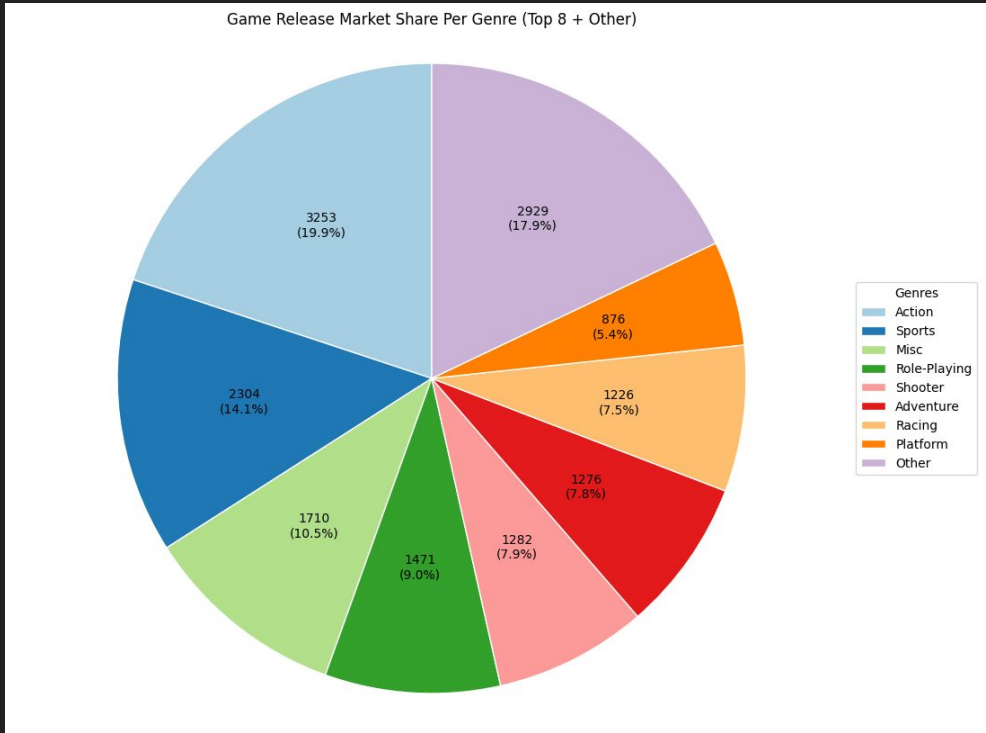
Market Share Per Genre

- The top five genres in the dataset are: Action, Sports, Misc (Miscellaneous), Role-Playing, and Shooter.

- Action games significantly outnumber Sports games (41%).

- Action games has dominant position in the market share.

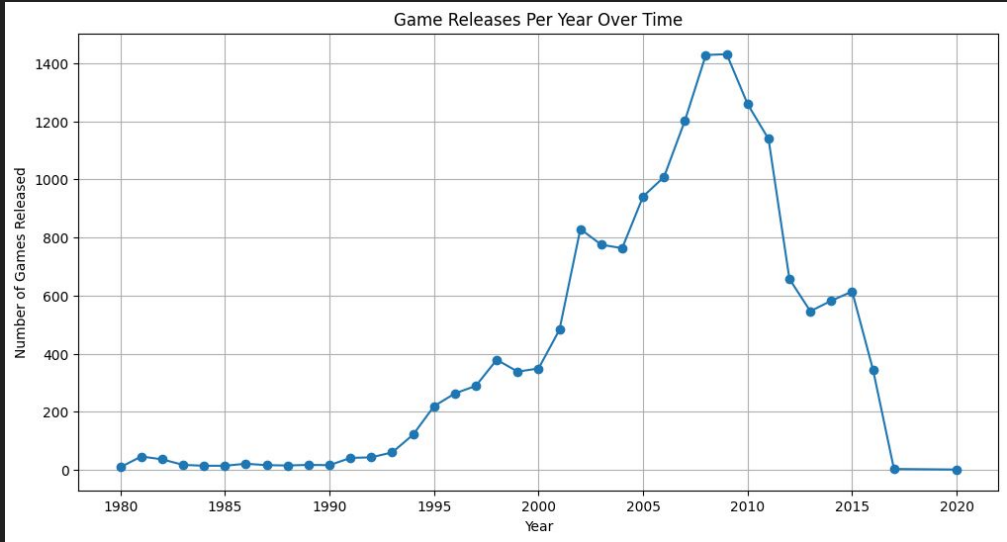
- certain degree of diversity.



Game Releases Per Year Over Time

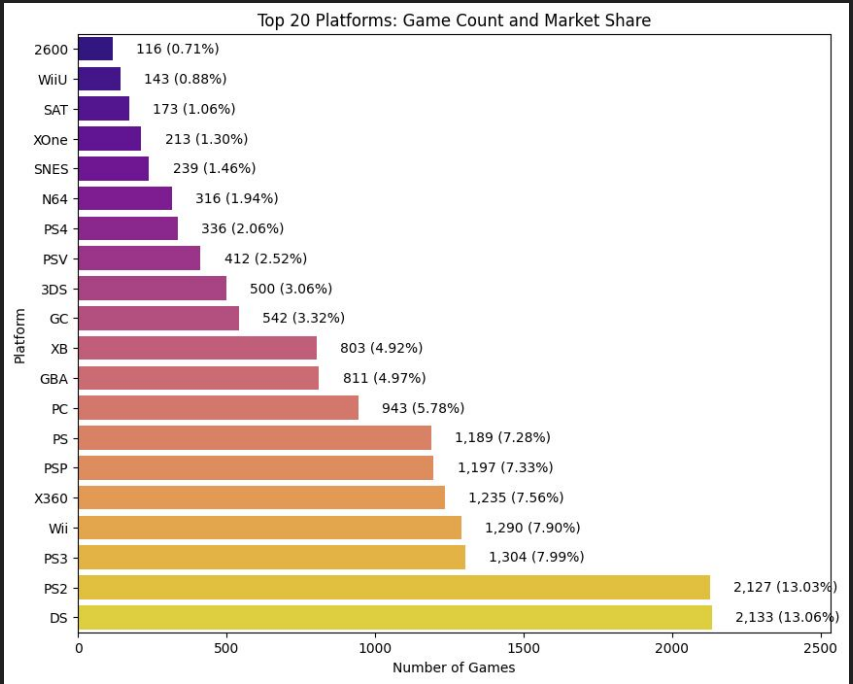
- The peak release period was concentrated between 2007 and 2011

- highest volume of releases (1,431 games)



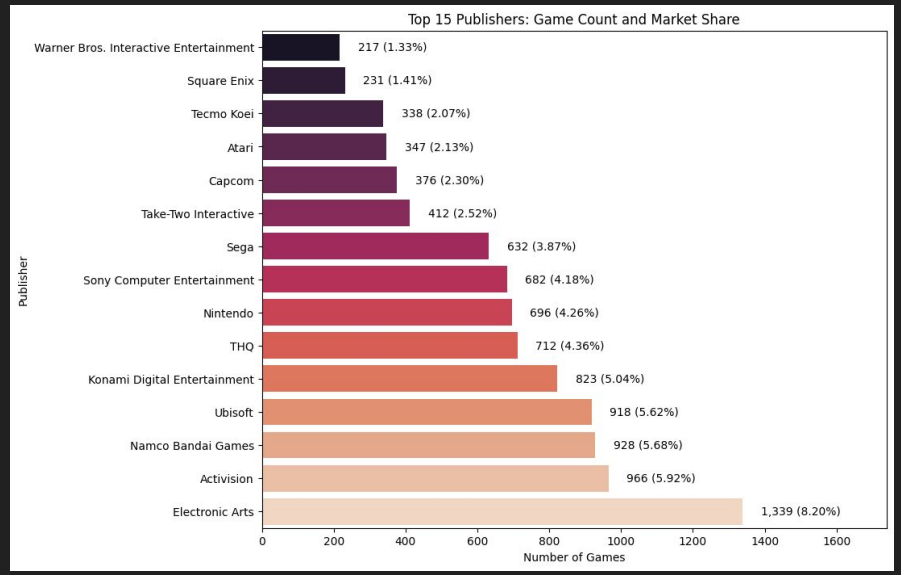
Top 20 Platforms

- Home consoles were the primary drivers
- DS stands out as a unique case study
- overlapping competitive period from 2005 to 2013



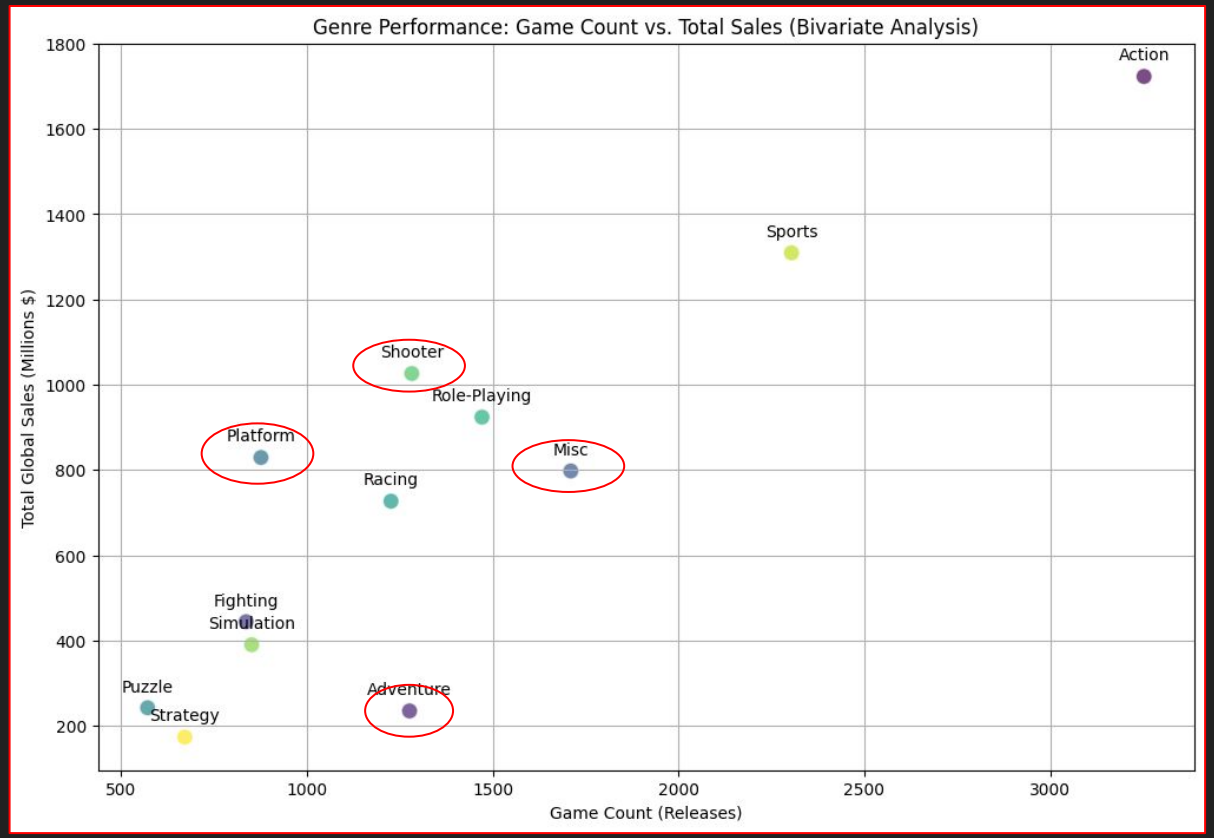
Top 15 Game Publishers

- highly concentrated among a few major publishers
- EA 1,339 games released (8.2% market share)
- Top three publishers (nearly 20%)



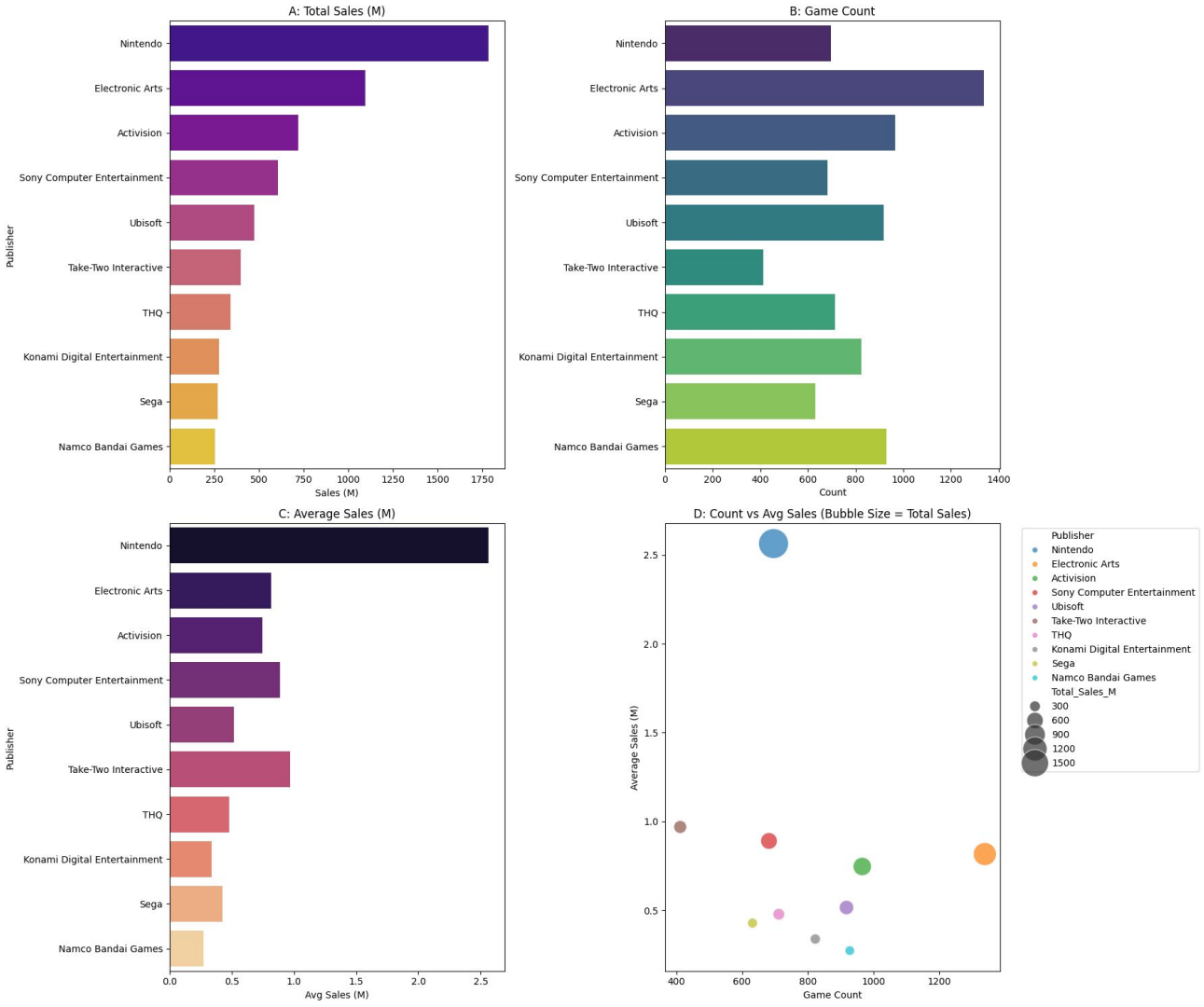
Bivariate Analysis

Total Global Sales



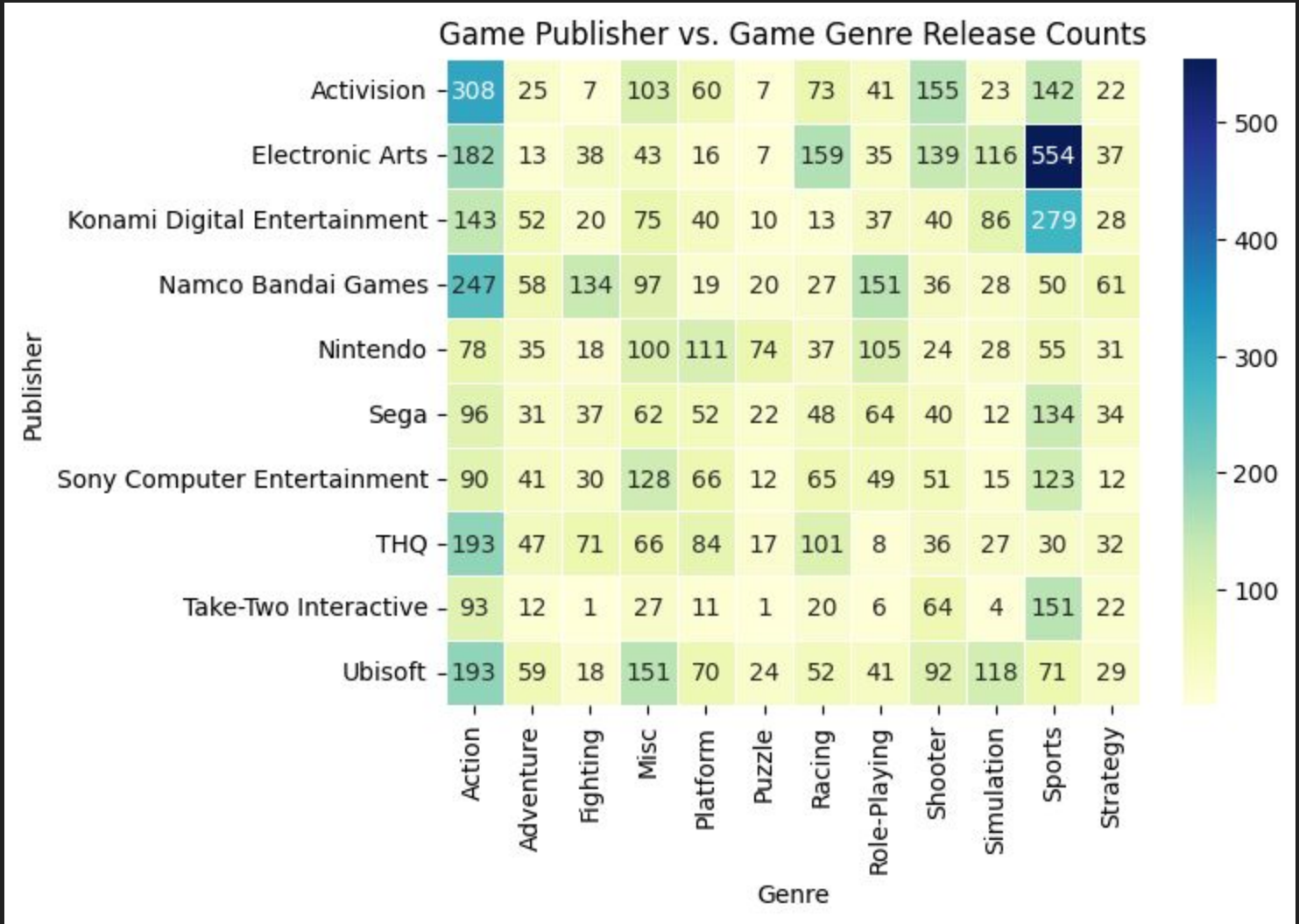
Total Global Sales

Game Genre	Count and Rank	Sales(in millions) and Rank
Misc	(3) 1710	(6) 797
Shooter	(5) 1282	(3) 1026
Adventure	(6) 1276	(11) 234
Planform	(8) 876	(5) 829


A: Total Sales (M)
Publisher
Nintendo
Electronic Arts
Activision
Sony Computer Entertainment
Ubisoft
Take-Two Interactive
THQ
Konami Digital Entertainment
Sega
Namco Bandai Games
0
250
500
750
1000
1250
1500
1750
Sales (M)
B: Game Count
Nintendo
Electronic Arts
Activision
Sony Computer Entertainment
Ubisoft
Take-Two Interactive
THQ
Konami Digital Entertainment
Sega
Namco Bandai Games
0
200
400
600
800
1000
1200
1400
Count
C: Average Sales (M)
Publisher
Nintendo
Electronic Arts
Activision
Sony Computer Entertainment
Ubisoft
Take-Two Interactive
THQ
Konami Digital Entertainment
Sega
Namco Bandai Games
0.0
0.5
1.0
1.5
2.0
2.5
Avg Sales (M)
D: Count vs Avg Sales (Bubble Size = Total Sales)
Average Sales (M)
0.5
1.0
1.5
2.0
2.5
400
600
800
1000
1200
Game Count
Publisher
Nintendo
Electronic Arts
Activision
Sony Computer Entertainment
Ubisoft
Take-Two Interactive
THQ
Konami Digital Entertainment
Sega
Namco Bandai Games
Total_Sales_M
300
600
900
1200
1500

Key Insights: Top 10 Publishers

- Nintendo is far ahead of second-place EA in terms of total unit sales
- EA and Activision have very high publication counts
- Take-Two has a relatively small publication count, but second-highest average sales per game
- Namco has a high volume of releases,but average sales per game are low.

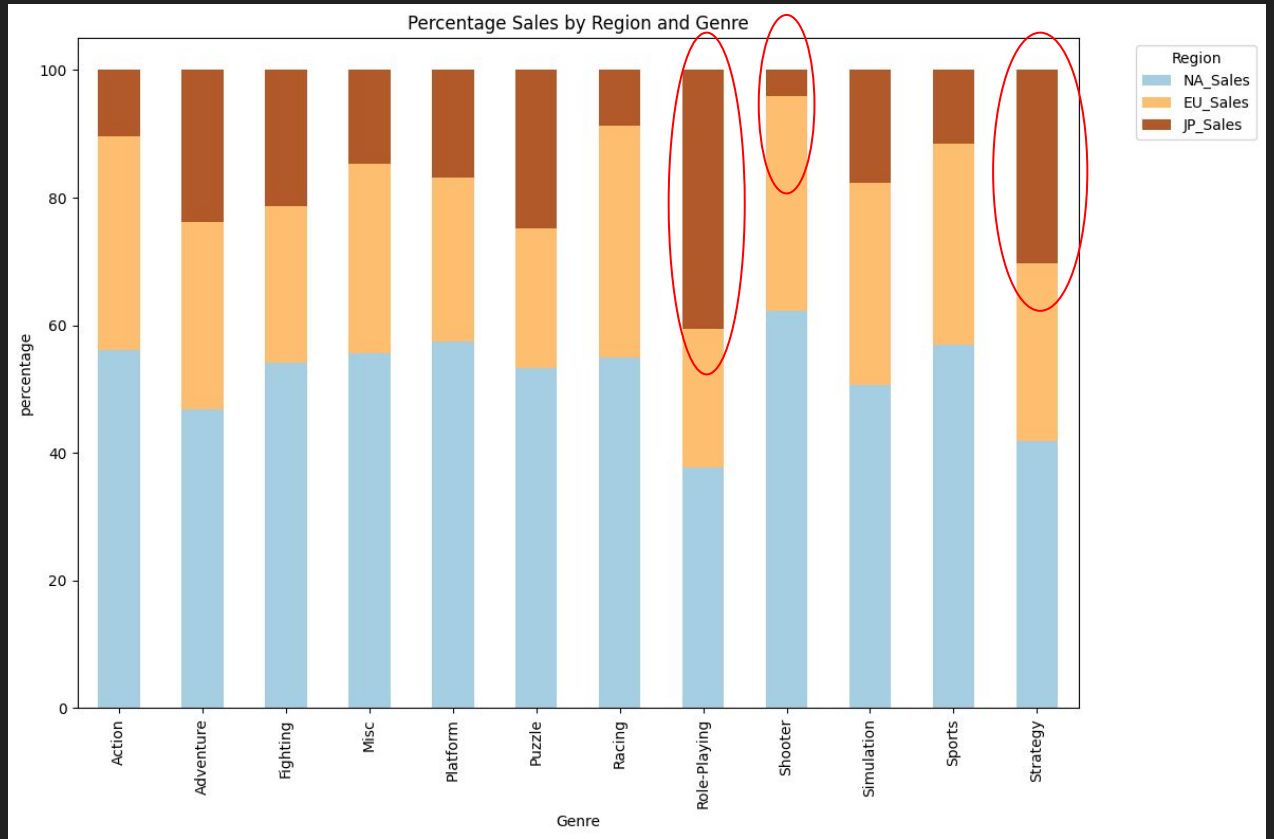

Game Publisher vs. Game Genre Release Counts
Publisher
Genre
Activision 308 25 7 103 60 7 73 41 155 23 142 22
Electronic Arts 182 13 38 43 16 7 159 35 139 116 554 37
Konami Digital Entertainment 143 52 20 75 40 10 13 37 40 86 279 28
Namco Bandai Games 247 58 134 97 19 20 27 151 36 28 50 61
Nintendo 78 35 18 100 111 74 37 105 24 28 55 31
Sega 96 31 37 62 52 22 48 64 40 12 134 34
Sony Computer Entertainment 90 41 30 128 66 12 65 49 51 15 123 12
THQ 193 47 71 66 84 17 101 8 36 27 30 32
Take-Two Interactive 93 12 1 27 11 1 20 6 64 4 151 22
Ubisoft 193 59 18 151 70 24 52 41 92 118 71 29
Action
Adventure
Fighting
Misc
Platform
Puzzle
Racing
Role-Playing
Shooter
Simulation
Sports
Strategy
500
400
300
200
100

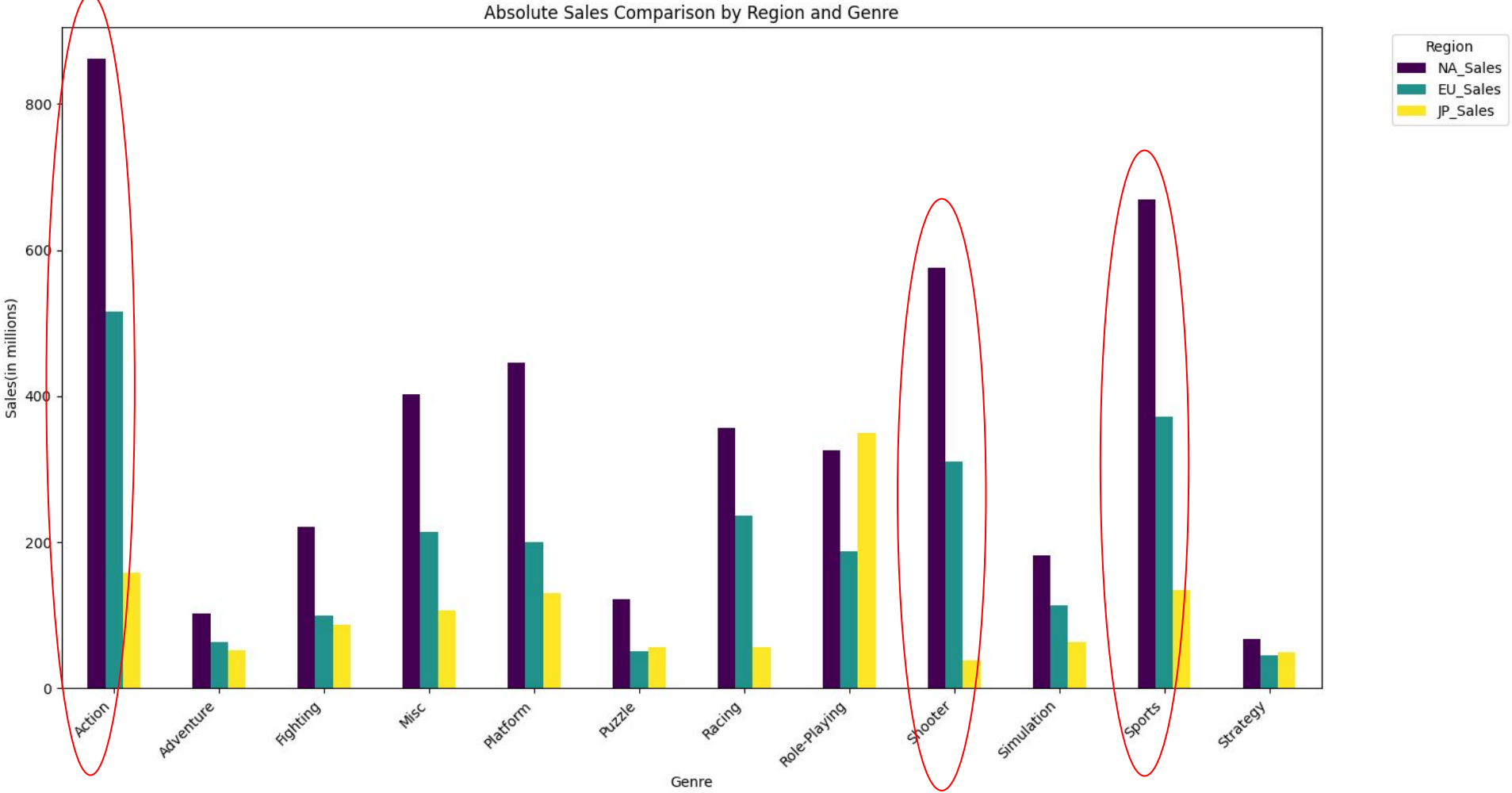
Publisher Strategies and Genre Distribution

- EA dominate the sports games market
- Activision dominate the shooter&action games market
- Nintendo dominate the Role-playing,Planform,Misc games market
- Most successful publishers concentrate their resources on the top six best-selling game genres
- Nintendo avoids the mainstream Action/Shooter markets
- Success in the gaming market is not achieved through a single, uniform model

Multivariate Analysis

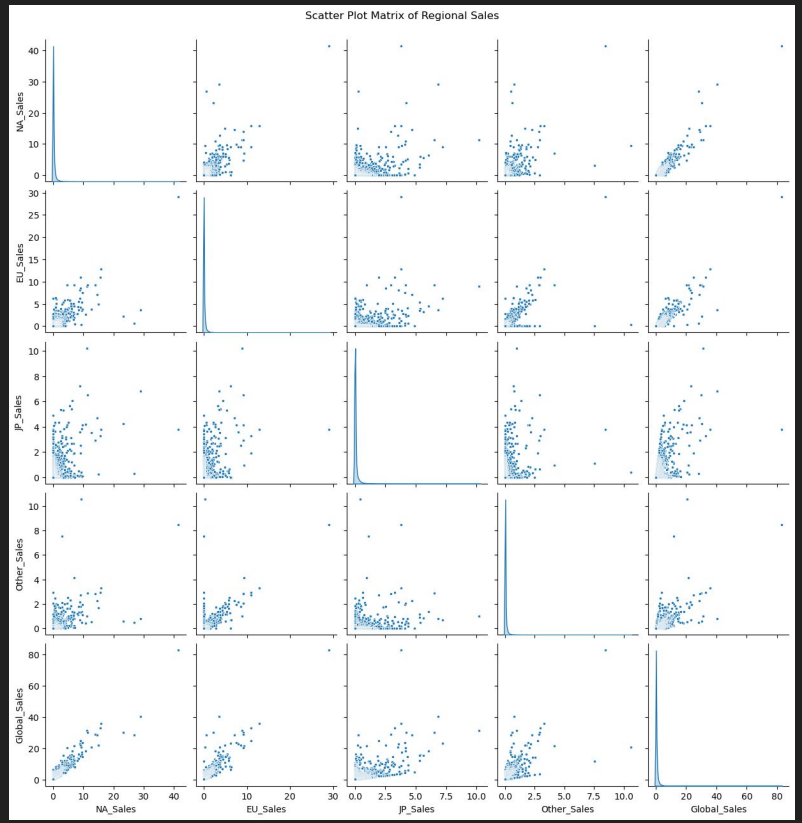
Percentage Sales by Region and Genre




Absolute Sales Comparison by Region and Genre
Region
NA_Sales
EU_Sales
JP_Sales
Sales(in millions)
0
200
400
600
800
Action
Adventure
Fighting
Misc
Platform
Puzzle
Racing
Role-Playing
Shooter
Simulation
Sports
Strategy
Genre

Cross-Regional Sales Trends

- majority of video games have low sales volumes
- North American and European markets share very similar tastes
- Relative Independence of the Japanese Market



Challenges and Limitations

- **Data Cleaning and Missing Values**
  -The Year and Publisher columns contained missing values (Null/NaN)
- **Data Timeliness**
  -The dataset only extends up to 2016.
- **Insufficient Granularity**
  -The data is aggregated by year, lacking specific months or dates
- **Single Data Source**
  -Data is sourced solely from kaggle
  -Didn’t include all game platform

Future Work

Developing a predictive model to forecast future global sales trends.

Merging External Data Sources

Limitations

Time Constraints: The project timeline was limited, preventing deep dives into complex predictive modeling.

Data Limitations: The lack of data granularity (e.g., no monthly data) made certain advanced analyses (like precise seasonality analysis) infeasible.

Scope of Project: The primary goal of this project was Exploratory Data Analysis (EDA), focusing on understanding existing data features rather than building a production-ready predictive model.

Q & A

Thank you!